Due Date: 17.11.2026, 23:59

CENG 113 - Programming Basics

Project 1 Weekly Task Assignment

Project overview

Develop a menu driven Python program for a software company with 50 employees in six departments. Your program will assign weekly tasks, generate department meeting reports, and allow each department manager to approve or reject their own department report after password verification.

Programming requirements

- Use meaningful variable names and brief comments where necessary.
- Do not use lists, tuples, dictionaries, sets, classes, external packages or imports. Do not use split(), readlines(), or list().
- You may use open(), with, readline(), write(), strip(), string concatenation, f-strings and basic conversions.

Input files

Keep employees.txt and manager_passwords.txt in the same directory as the Python program. Both files must remain unchanged.

Employee records

employees.txt contains 50 records. Each record contains the following six nonempty lines in this exact order: Employee ID, Name, Surname, Title, Department, Role. There is exactly one empty line between consecutive records.

Sample records

1001
Deniz
Kaya
Software Development Manager
Software Development
Manager

1002
Ayse
Yilmaz
Backend Developer
Software Development
Employee

Employee IDs run from 1001 through 1050. Role is exactly Manager or Employee. Titles and department names may contain spaces; keep each complete line as one value.

Department	Department Name	Employees	Manager ID
1	Software Development	10	1001
2	Artificial Intelligence	9	1011
3	Research and Development	8	1020
4	Quality Assurance	8	1028
5	IT Operations	8	1036
6	Human Resources	7	1044

Manager password records
manager_passwords.txt contains six records. Each record has two nonempty lines: Manager ID followed by Password. There is exactly one empty line between consecutive records. Password matching is case sensitive.

Sample records

1001
Team1Week26

1011
Team2Week26

Plain text password storage is a teaching simplification; real systems require secure password handling.

Main menu
1. Display department employees

2. Assign weekly tasks

3. Generate meeting report

4. Display meeting report

5. Review meeting report

6. Display task summary

0. Exit

After options 1 to 6, return to the main menu. Reject invalid choices and prompt again.

Required functions
Implement the following functions. Add helper functions where useful. Functions that display information or save files do not need to return a value.

Function	Purpose or return value
display_menu()	Display the main menu.
select_department()	Validate and return a choice from 1 to 6.

Function	Purpose or return value
display_employees(department)	Display employees in the selected department.
assign_weekly_tasks(department)	Collect and save weekly tasks. Allow replacement of existing assignments only after report rejection.
generate_meeting_report(department)	Collect missing meeting metadata and build the report.
display_meeting_report(department)	Display the report, or explain that none exists.
authenticate_manager(manager_id, password, department)	Return True only when all authentication checks pass.
review_meeting_report(department)	Handle login attempts, decision input and report updates.
display_task_summary(department)	Calculate and display counts, total hours and average.
main()	Run the menu loop and call the other functions.

The department parameter is the numeric choice from 1 to 6. Map it to the exact department name.

Menu operations

1 Display department employees
Select a department and display its employees in their original file order, including its manager. Show each employee ID, full name and title.

2 Assign weekly tasks
Process the employees of the selected department in their original file order, including its manager. Display each employee ID, full name and title. Ask for one nonempty task description and an integer number of estimated hours from 1 to 40. If the task description is empty or the estimated hours are invalid, display an appropriate error message and ask the user to enter the value again.

Assign one task to every employee in the department. Save the completed assignments to tasks_N.txt, where N is the department number from 1 to 6. For example, tasks_1.txt belongs to Software Development. Do not modify another department's files.

Task file format
Each task record contains three nonempty lines: Employee ID, Task description, Estimated hours. Separate records with one blank line. Do not include headers.

```
1001
Review the weekly development plan
6

1002
Implement the user profile endpoint
12
```

The program manages tasks for a single week. If no assignments exist for the selected department, collect and save tasks for all employees. Existing assignments may be replaced only if the department's

meeting report has been rejected. In this case, ask for confirmation before collecting new tasks and overwriting tasks_N.txt. If the user declines, return to the main menu without making any changes.

Assigning weekly tasks does not create or update the meeting report. After the initial assignment or reassignment following a rejection, the user must select Generate Meeting Report (option 3). This option creates the report or overwrites the rejected report using the saved assignments, sets its status to Pending Approval, and clears the previous review details.

3 Generate meeting report

When generating a report for the first time, ask for a nonempty meeting date and a nonempty one line meeting note. A date such as 2026-11-02 is acceptable; calendar validation is not required. A report can be generated only when every employee in that department has exactly one valid task.

If the report status is Pending Approval or Approved, inform the user that the report already exists and return to the main menu without changing any files. If the report is Rejected, ask for confirmation before overwriting it using the currently saved task assignments. Set the status to Pending Approval and clear the previous review details.

Persistent review information

Use review_N.txt to preserve report metadata and review status between program runs. It contains exactly six nonempty lines in this order. This is one record, so it needs no blank separator.

Line	Field	Initial value
1	Status	Pending Approval
2	Reviewer ID	-
3	Reviewer full name	-
4	Review comment	-
5	Meeting date	Entered by user
6	Meeting note	Entered by user

Create review_N.txt when the meeting report is first generated. When regenerating a rejected report, preserve the meeting date and note, set the status to Pending Approval, and reset the reviewer ID, reviewer name and review comment to a hyphen.

Report content

Write the human readable report to meeting_report_N.txt. Include the department, meeting date, meeting note, status, and each employee's ID, full name, title, task and estimated hours. End with the number of employees, total estimated hours and average estimated hours to two decimal places. Include the reviewer ID, reviewer name and review comment, using a hyphen when no review exists.

4 Display meeting report

Select a department and display the contents of meeting_report_N.txt. If no report exists, inform the user and return to the main menu. Displaying a report must not change its contents or status.

5 Review meeting report

Only reports with the status Pending Approval can be reviewed. If the report is already Approved or Rejected, display its current status and return to the main menu.

Require an existing report before asking for credentials. Prompt for a manager ID and password. Allow at most three attempts during each use of option 5. A login succeeds only if all of the following conditions hold:

- The ID exists in employees.txt and its role is Manager.
- The manager belongs to the selected department.
- The ID and password match a record in manager_passwords.txt.

On failure, display: Invalid manager ID or password, or unauthorized department. After three failures, return to the main menu. A valid manager from another department must not be allowed to review the selected report. Never write passwords to reports or output files.

After successful authentication, display the report and offer Approve, Reject and Cancel. An approval comment is optional; store a hyphen if it is blank. A rejection comment is required and must not be empty. Cancel preserves the current decision.

Save Approved or Rejected, reviewer ID, reviewer full name and review comment in review_N.txt. Preserve the meeting date and note. Rebuild meeting_report_N.txt with the new decision.

6 Display task summary

Option 6 displays the department employee count, number of assigned tasks, total estimated hours and average hours per assigned task. Before any tasks exist, show zero assigned tasks and zero total hours; display N/A for the average.

0 Exit

Terminate the program when the user selects option 0.

Validation and exception handling

- Reject invalid menu and department choices. Keep asking until a valid choice is entered.
- Handle nonnumeric hour input with try/except ValueError. Enforce the inclusive 1 to 40 range.
- Reject blank task descriptions, meeting dates, meeting notes and rejection comments.
- Handle missing files with FileNotFoundError. Treat absent generated files as an expected initial condition.
- Ensure that all files are closed after use.
- Read saved tasks and review metadata when the program starts again. Previously saved assignments and decisions must remain available.

Suggested checks before submission

- Display all employees of one department.
- Try invalid menu and department choices, blank task descriptions, blank meeting dates and notes, nonnumeric hour inputs, and hour values outside the allowed range of 1 to 40. Verify that the program rejects these inputs and asks for valid values.
- Try to generate a report before tasks exist and try to review a report before it exists.
- Assign tasks to a department, generate its report and verify employee details, counts, totals and average.

- Try an unknown ID, a wrong password and another department's valid manager credentials. Verify the three attempt limit.
- Reject the report with a comment, then verify the updated report. Confirm that a blank rejection comment is refused.
- Approve a report with the correct manager credentials. Confirm that the decision survives program exit and restart.
- Reassign tasks after rejection, then select Generate Meeting Report. Verify that the report contains the revised assignments, its status is Pending Approval, and the previous review details have been cleared.
- Try an operation that requires a missing input file. Verify that the program displays an appropriate message and returns to the main menu.
- Try to regenerate a Pending Approval or Approved report and replace its task assignments. Verify that the existing files remain unchanged.
- Regenerate a rejected report without reassigning tasks. Verify that the saved assignments are used, the status becomes Pending Approval, and the previous review details are cleared.

Project demonstration

Before the presentation, prepare sample task and report files using your program. You may use these files during the demonstration; you do not need to enter tasks for an entire department live. During the presentation, you will be asked to demonstrate selected operations and explain the relevant parts of your code. Every member must be able to explain the complete program. Your program must support all required features, although not all of them will be demonstrated.

Submission Rules

1. Submit your Python source file in the format of **CENG113_(group_number)_Project1.py (e.g., CENG113_G1_Project1.py)** with sample generated task, review and meeting report files for at least one department. Write your student IDs on top as comment in the Python file.
2. Only one student should submit the project on behalf of the group. Each group must make **only one submission**.
3. Please add **clear, meaningful comments** to your code to explain what each part does. Submissions without sufficient comments will result in point deductions.

No plagiarism. Submissions must be your group's own work. You are not allowed to share code between different groups.